

Reconstructing Value Chains from FactSet Data: An Exploratory Analysis Using Network Science

Internship Report — Master's Degree in Physics, Curriculum "Physics of Complex Systems and Big Data", University of Rome "Tor Vergata"

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Abstract

This study presents an exploratory topological analysis of the global value chain constructed by integrating firm-level supply chain data with product-level revenue classifications (RBICS) provided by FactSet. Moving beyond traditional coarse-grained macroeconomic models, this work employs a network science framework to project corporate commercial interactions into a granular product-product space. The resulting 2018 empirical value chain exhibits a densely entangled topology characterized by a macroscopic density of 12.9%. Statistical characterization reveals broadly distributed, right-skewed profiles for both node degrees and link weights bounded by sharp structural cutoffs, indicating that the network is heavily populated by weak, potentially spurious connections generated by the bipartite projection of diversified corporate portfolios. Despite this pervasive structural noise, a temporal analysis spanning from 2018 to 2023 demonstrates significant macroscopic stability, with persistent links exhibiting distinctively higher transition probabilities compared to transient connections. These findings highlight the presence of a robust economic backbone obscured by topological artifacts. Consequently, this report establishes the methodological necessity of implementing a maximum-entropy null model (DBCM) to statistically filter the network and isolate the genuine functional dependencies of the global economy.

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1. Introduction

In recent years, the application of statistical mechanics and network science to economic complexity has provided powerful tools for understanding global markets. Traditionally, macroeconomic flows have been analyzed using Input-Output (I-O) tables, which map the monetary interdependencies between different economic sectors [2]. However, I-O tables suffer from a severe "coarse-graining" problem: by aggregating the economy into a few dozen macro-industries, they inevitably average out and obscure the microscopic dynamics of the thousands of individual products actually exchanged in the market. Indeed, recent literature has increasingly pointed out the inadequacy of coarse-grained, industry-level data for the detailed study of supply chains and production networks [6].

To overcome this limitation, econophysics treats economic entities as interacting particles within a complex network. In this framework, statistical mechanics provides rigorous methods to distinguish genuine structural patterns from random fluctuations. Specifically, as formalized in the literature [1], this is achieved by constructing maximum-entropy null models, which mathematically simulate microcanonical or canonical ensembles of randomized networks. These ensembles preserve specific local constraints of the original empirical network—such as the degree sequence—while randomizing all other connections. This approach allows isolating statistically significant interactions, filtering out the intrinsic noise of real-world complex systems and revealing their true structural backbone.

The primary objective of this study is to apply the network science mathematical framework

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to proprietary financial data provided by FactSet, moving from a firm-level perspective to a product-level paradigm. Specifically, the study leverages two main datasets: a binary "firm-firm" supply chain network, which maps the direct commercial buyer-supplier relationships between companies, and the RBICS (Revere Business Industry Classification System) dataset, which categorizes the specific production lines of each firm. By merging the firm-level supply chain with the RBICS product classification, it is possible to reconstruct an empirical "value chain". Unlike a simple supply chain, a value chain focuses on the value added at every stage of the production process, tracking the directional transformation of goods from raw materials (basic inputs) to final consumer outputs [6].

This report details the construction of a product-level value chain from these FactSet datasets and presents a preliminary topological exploration of the resulting network, laying the necessary groundwork for future advanced statistical validations. The document is structured as follows. Following this introduction, Section 2 provides an overview of the FactSet datasets, detailing their architecture and original data collection methodology. Section 3 describes the data preprocessing and cleaning procedures required to handle missing or inconsistent values. Section 4 outlines the mathematical and operational logic behind the value chain construction, explaining the merging process between firm and product data. Section 5 presents an exploratory topological analysis of the raw value chain obtained for the year 2018, focusing on weight distributions, degree distributions, major hubs, and the identification of boundary nodes representing basic and final products. Section 6 extends the scope to a temporal persistence analysis from 2018 to 2023, utilizing Jaccard similarity matrices to construct and evaluate a persistent network over time. Section 7 summarizes and discusses the conclusions drawn from this preliminary work, highlighting the topological complexity and the structural noise present in the raw data, thereby addressing the theoretical necessity of implementing a statistical null model.

2. FactSet Data Overview

The empirical foundation of this study relies on two proprietary datasets provided by FactSet: the Supply Chain Relationships database and the Revere Business Industry Classification System

(RBICS). Together, these datasets provide the necessary microscopic information to map both the commercial interactions between firms and their specific productive outputs.

2.1. FactSet Supply Chain Relationships

The FactSet Supply Chain dataset exposes and categorizes the complex business interconnections among global companies. The data collection process is deeply rooted in regulatory disclosures. Under US accounting standards, specifically FASB ASC 280 (formerly SFAS 131), publicly traded companies are legally required to disclose the identities of major customers that account for 10% or more of their total consolidated revenues [7]. Crucially, FactSet does not only record "direct" relationships explicitly declared by a reporting company; they also construct "reverse" links to non-disclosing parties. For example, if Company A reports Company B as a major customer, FactSet automatically infers and records a supplier relationship from A to B [5].

In the context of this study, this dataset provides key variables such as the unique identifiers for suppliers and customers, alongside ultimate parent entity mappings and the temporal validity of the commercial ties. From a network science perspective, this structure translates into a directed and binary "firm-firm" supply chain network, where nodes represent individual companies and the directed edges map the flow of goods and services from suppliers to buyers. As illustrated in Figure 1a, the raw structure is stored as a directed edgelist, which is subsequently mapped into the macroscopic directed network topology shown in Figure 1b. The raw supply chain network for the baseline year 2018 comprises 86,000 corporate nodes and 306,539 directed edges. This structural configuration yields an extremely sparse topology, characterized by a density of $\rho = 0.000041$ and an average degree of $\hat{k} = 7.13$.

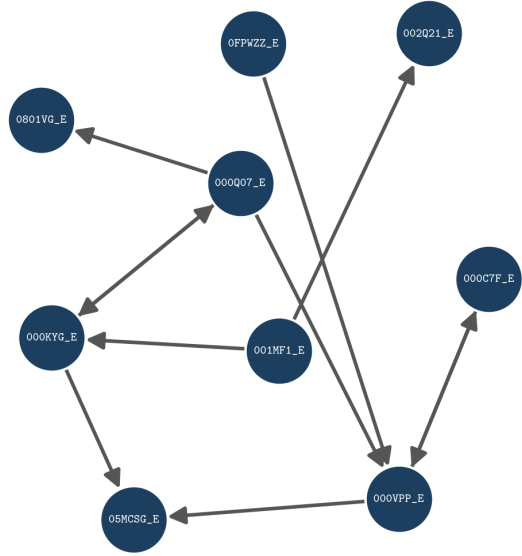
2.2. FactSet RBICS Revenue Dataset

While the Supply Chain dataset maps *who trades with whom*, RBICS provides the essential information on *what is being produced*. This study utilizes the granular RBICS Revenue dataset.

RBICS is a comprehensive, structured taxonomy designed to classify global companies based on their operating footprints. The taxonomy employs a bottom-up classification approach arranged

Suppliers	Customers
000Q07_E	0801VG_E
000Q07_E	000KYG_E
000Q07_E	000VPP_E
⋮	⋮
001MF1_E	000KYG_E
000KYG_E	05MCSG_E
000KYG_E	000Q07_E

(a) Raw edgelist excerpt



(b) Topological sketch

Figure 1: **Topological sample of the Supply Chain network (2018)**. In (a), an excerpt of the raw directed edgelist modeling the commercial relationships for the year 2018. In (b), a topological sketch illustrating how these binary ties interconnect to form the macroscopic directed network. Crucially, representing these entities as nodes in a complex network enables comprehensive topological analyses; for instance, it allows us to structurally trace how a single supplier, such as 000Q07_E, connects and distributes value to multiple downstream targets (e.g., 0801VG_E, 000KYG_E, and 000VPP_E).

in a 14×6 matrix, featuring six hierarchical levels¹. These range from broad macro-economies at Level 1 (e.g., "Industrials" or "Healthcare") down to highly specific sub-industries at Level 6. Strictly speaking, within the RBICS taxonomy, these Level 6 categories represent granular commercial sectors at a product level (e.g., "Alumina and Aluminum Products Manufacturing" or "Surgical Robotic Systems") rather than individual physical goods. However, for the sake of brevity and clarity, we will refer to these Level 6 commercial sectors simply as "products" throughout the remainder of this paper. The RBICS Revenue dataset unpacks a company's total revenue stream, mapping specific business segments to their exact revenue percentages and assigning them to the corresponding categories [4]. As illustrated by the raw edgelist sample in Figure 2a and its corresponding bipartite network

¹For a comprehensive overview of the taxonomy structure and the 12 anchor industries, see Appendix A and [3].

representation in Figure 2b, a single firm's revenues might be distributed across multiple distinct products.

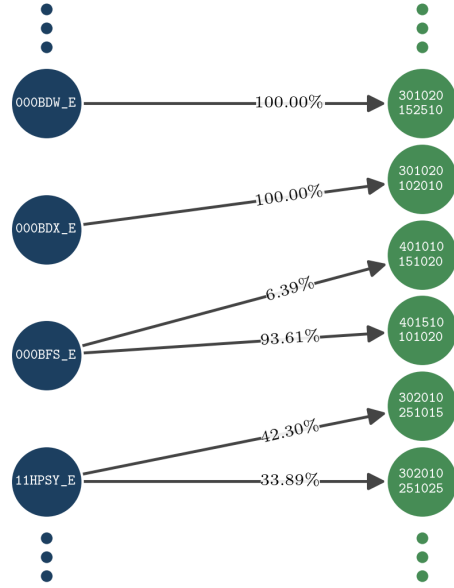
For the purposes of network construction, RBICS acts as a directed and weighted bipartite network mapping firms to their respective products. The edges are directed from the firm to the product, and the weights are the revenue percentages generated by the firm in that specific sector. To illustrate the macroscopic scale of these original data, the raw bipartite network for the reference year 2018 comprises 72,365 nodes and 131,380 directed edges, exhibiting an extreme sparsity with a density of $\rho = 0.000025$ and an average degree of $\hat{k} = 3.63$.

2.3. Temporal Coverage and Dataset Integration

The data extracted for this exploratory analysis spans a six-year temporal window, with annual snapshots available from 2018 to 2023. This longitudinal structure enables the study of the temporal

Firms	Products	Weight
000BDW_E	301020152510	100.00%
000BDX_E	301020102010	100.00%
000BFS_E	401010151020	6.39%
⋮	⋮	⋮
11HPSY_E	302010251015	42.30%
11HPSY_E	302010251025	33.89%
11JYLZ_E	351020651010	100.00%

(a) Raw edgelist excerpt



(b) Topological sketch

Figure 2: **Topological sample of the bipartite RBICS network (2018).** In (a), a sample of the raw weighted edgelist derived from the FactSet RBICS Revenue dataset. In (b), its exact topological representation as a directed, weighted bipartite network. The left nodes represent corporate entities, whereas the right nodes correspond to the identifying codes of the production sectors at the highest level of granularity (L6-level). The edge weights reflect the exact percentage of revenue that a company derives from that specific sector. The graph highlights distinct production strategies: "pure" or single-product companies, such as 000BDW_E, which derives 100% of its revenue from the sector 301020152510 (Banking Operations); and diversified companies, such as 000BFS_E, whose revenues are split between 401010151020 (Defense Contractors, 6.39%) and 401510101020 (Fuel, Component and Maintenance Providers, 93.61%).

persistence of the economic networks (as will be discussed in Section 6).

However, leveraging these raw datasets poses significant analytical challenges. Because they are independently compiled and cover overlapping yet distinct market segments, an inherent alignment problem emerges. Although both datasets utilize the same firm identifier codes, not all companies present in the firm-firm supply chain network are covered by the RBICS revenue database; conversely, not all categorized RBICS companies disclose their supply chain ties. This structural misalignment, coupled with the inevitable presence of missing values and classification anomalies, dictates that the raw datasets cannot be naively merged. A rigorous and systematic data preprocessing and cleaning phase is strictly required before any valid value chain network can be constructed. These pro-

cedures will be detailed in the following section.

3. Data Pre-processing

Raw financial datasets are inherently prone to noise, redundancies, and missing entries, making a rigorous preprocessing pipeline a mandatory step before any robust network construction can occur. To prepare the FactSet datasets for topological analysis, we implemented a systematic cleaning procedure to filter out uninformative data and isolate the variables strictly necessary for modeling the economic interactions.

To optimize computational efficiency and focus solely on the structural properties of the networks, both datasets were stripped down to their essential features.

For the binary firm-firm supply chain network, the retained information was strictly limited to the operational entity identifiers² of the supplier and the customer, defining the unweighted directed edges between corporate nodes.

Conversely, for the bipartite RBICS firm-product network, the retained columns included the unique firm entity identifier, the specific product code, the product name, and the associated revenue percentage. This study strictly maintains the product classification at the L6 level of granularity, which is the most detailed hierarchical tier available in the RBICS taxonomy, ensuring a truly microscopic description of the value chain.

Following the feature selection, a noise reduction phase was executed to handle missing and mathematically inconsistent data. Any records containing missing values in the critical mapping columns were entirely dropped from the analysis. While the unweighted supply chain dataset was not affected by this specific step, this filtering was particularly crucial for the RBICS dataset, where any entry reporting a revenue percentage less than or equal to zero was discarded. From a physical and network modeling perspective, a negative or null revenue weight carries no structural meaning and acts merely as statistical noise that could distort subsequent topological metrics.

A fundamental preprocessing step was applied exclusively to the RBICS dataset to ensure mathematical consistency in the bipartite network weights. Due to standard financial reporting practices, a single corporate entity often disaggregates its revenues by geographic or operational divisions, causing multiple internal business segments to map to the exact same L6 product code. To prevent the creation of duplicate edges, these parallel revenue streams were grouped and aggregated by firm and by L6 product by simply summing their revenue percentages.

²Rather than aggregating commercial ties under the highest-level consolidating holding company within a corporate hierarchy (the "ultimate parent entity" or "ultimate owners"), this study maps relationships at the operational entity level using FactSet entity identifiers. An operational entity represents a distinct legal or functional subsidiary, plant, or division that actively engages in commercial transactions and physical production flows. This granular approach prevents the over-consolidation of diverse business units into single parent entities, thereby preserving the true micro-topology of the supply chain and capturing localized operational bottlenecks and structural pathways that macro-level holding consolidation would otherwise obscure.

However, due to corporate reporting anomalies or the previous removal of non-positive values, the aggregate revenue percentages for a single firm might not necessarily sum to exactly 100%. To resolve this and ensure that the out-strength of every firm node is properly standardized, the edge weights were normalized. Each aggregated revenue percentage was divided by the firm's new total valid revenue and multiplied by 100. This normalization guarantees that the outgoing weighted edges from any given firm act as a coherent probability distribution of its production, always summing to exactly 100%.

The output of this entire preprocessing pipeline yields clean, standardized, and interoperable edge lists, ready to be merged into the unified value chain structure described in the next section. The cleaned firm-firm supply chain network still comprises 86,000 corporate nodes and 306,539 directed edges, with a density of $\rho = 0.000041$ and an average degree of $\hat{k} = 7.13$. Concurrently, the cleaned bipartite firm-product network consists of 20,323 nodes connected by 48,584 weighted edges, exhibiting a density of $\rho = 0.000118$ and an average degree of $\hat{k} = 4.78$. These macroscopic metrics highlight the vast and sparse nature of the empirical data prior to its mathematical projection into the product space.

4. Value Chain Construction: The Merging Process

Once the datasets have been rigorously cleaned, the next methodological step consists of merging these networks to project the firm-level data onto the product space. In this projected space, each node represents a specific product, uniquely identified by its 12-digit L6 code. As previously mentioned, the fundamental prerequisite for this projection is finding the intersection of the two original datasets: only the firms present in both the Supply Chain and the RBICS network are retained. Without this overlap, it would be impossible to map the commercial ties to their corresponding productive outputs.

The logic underlying the projection relies on combining two distinct layers of information. From the supply chain network, we extract the explicit commercial relationships between firms (i.e., who supplies whom). Concurrently, from the RBICS bipartite network, we extract the specific products manufactured by each firm, along with their respective revenue percentages.

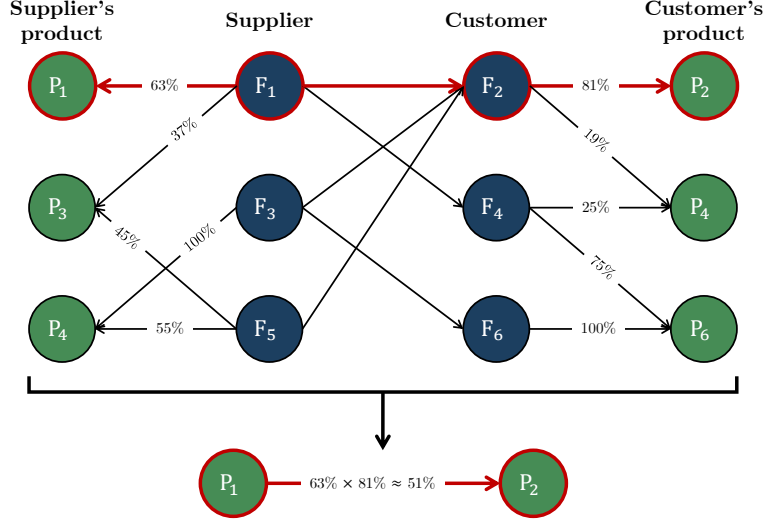


Figure 3: **Schematic representation of the product-product link inference.** The figure illustrates the methodology to project firm-level data onto the product space. The upper section integrates both datasets: the central nodes depict the firm-firm supply chain network (blue), highlighting a directed commercial tie from supplier F_1 to customer F_2 . The outer nodes (green) represent the bipartite RBICS mappings, where directed edges and their weights indicate the revenue percentage a firm derives from a specific product. The red highlighted path traces the flow from the supplier’s product (P_1) through the commercial tie to the customer’s product (P_2). To infer the final macroscopic value chain link $P_1 \rightarrow P_2$ (bottom section), we assume that the supplier’s output acts as an input for the customer’s production. The strength of this inferred link is estimated by multiplying the corresponding revenue shares along the path (e.g., $63\% \times 81\% \approx 51\%$).

To illustrate how a link between two arbitrary products, p_i and p_j , is inferred, let us consider a simplified scenario where product p_i is exclusively manufactured by firm f , and product p_j is exclusively manufactured by firm f' (the generalization to multiple firms will follow). We investigate the potential connection between these products by examining the supply chain for a commercial tie between their respective producing firms. If a directed link $f \rightarrow f'$ exists—indicating that f is a supplier to f' —it is economically reasonable to assume that firm f' purchases product p_i from firm f . Since firm f' is engaged in producing p_j , we can deduce a direct commercial dependency: p_i acts as a necessary input for the production of p_j . This establishes a directed product-product link, denoted as $p_i \rightarrow p_j$.

However, establishing the existence of a link is insufficient for a comprehensive value chain analysis; evaluating the structural strength of this connection requires assigning a specific weight. Since the original firm-firm supply chain network is binary (unweighted), the necessary weights must be

derived from the bipartite, weighted RBICS network. To quantify the strength of the inferred link $p_i \rightarrow p_j$, we extract the original revenue weights of the firm-product links ($f \rightarrow p_i$ and $f' \rightarrow p_j$) and compute their product. This multiplication yields a proxy for the probability that a transaction between the two firms actually corresponds to a flow from sector p_i to sector p_j . A visual example of this specific procedure is depicted in Figure 3.

Naturally, assuming that a specific pair of products (p_i, p_j) emerges from only a single pair of firms is highly restrictive. In the general empirical case, the same product-product interaction is generated by multiple distinct firm-firm commercial ties across the network. Mathematically, the weight $w_{p_i p_j}$ for the macroscopic link $p_i \rightarrow p_j$ is computed by summing the contributions over all firm pairs (f, f')

$$w_{p_i p_j} = \sum_{f, f'} a_{f, f'} w_{f, p_i} w_{f', p_j} \quad (1)$$

where $a_{f,f'}$ represents the element of the firm-level adjacency matrix ($a_{f,f'} = 1$ if a direct commercial tie exists from firm f to firm f' , and 0 otherwise). The terms w_{f,p_i} and w_{f',p_j} denote the specific weights connecting the respective firms to those products. To account for the multiplicity of these connections and extract a representative interaction, this total sum is subsequently normalized by the number of active firm-level transactions N_{p_i,p_j} that specifically map to the product pair (p_i, p_j) . This yields the average link weight $\langle w_{p_i p_j} \rangle$:

$$\langle w_{p_i p_j} \rangle = \frac{1}{N_{p_i, p_j}} \sum_{f, f'} a_{f, f'} w_{f, p_i} w_{f', p_j} \quad (2)$$

This averaging procedure ensures that the final product space edge reflects the typical, normalized strength of the commercial dependency across the entire economy.

Based on the theoretical framework discussed above, the algorithmic methodology implemented to project the raw data into the final product-product value chain can be summarized in the following sequential steps:

1. **Data Intersection:** Identify all active commercial ties ($f \rightarrow f'$) in the supply chain where both the supplier and the customer possess a valid, normalized product portfolio within the RBICS dataset.
2. **Cartesian Projection:** For every valid firm-level tie, generate all possible pairwise combinations between the supplier's output products (p_i) and the customer's output products (p_j). This mathematical projection establishes the topological existence of the inferred dependencies ($p_i \rightarrow p_j$).
3. **Probabilistic Weighting:** Assign an intermediate statistical weight to each newly generated product pair. This is achieved by computing the product of the supplier's revenue share for p_i and the customer's revenue share for p_j .
4. **Macroscopic Aggregation:** Since the exact same commercial dependency ($p_i \rightarrow p_j$) typically emerges from multiple independent firm-level transactions across the market, all individual intermediate weights for that specific pair are summed. This total is then divided by the number of distinct firm-to-firm transactions that map to this specific product pair, yielding the final averaged weight for the macroscopic value chain edge.

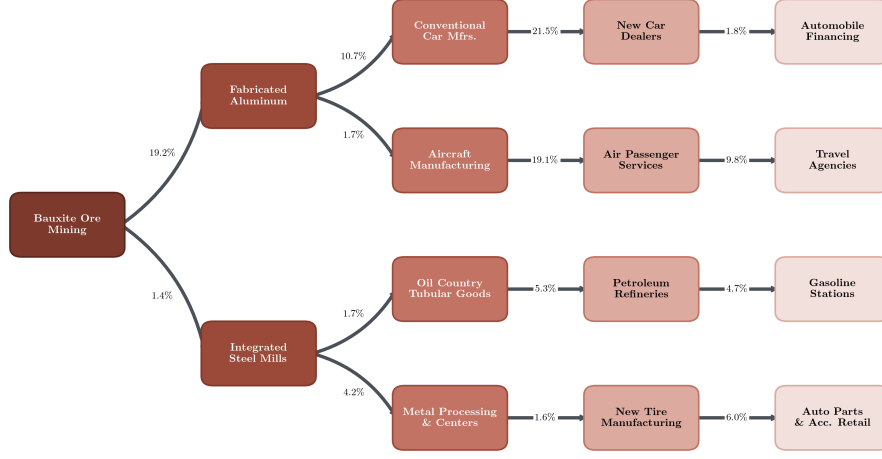
5. Exploratory Analysis of the 2018 Raw Network

The resulting value chain is represented by a directed network among products that maps the complex interconnectedness of the global market. The macroscopic network comprises 1,708 nodes (unique products) and 376,383 directed edges (inferred commercial dependencies), resulting in a high network density of $\rho = 12.9\%$ with an average degree of $\hat{k} = 220.4$. This dense topology indicates a significant degree of structural overlap, meaning that the true commercial backbone struggles to emerge from the background noise. This phenomenon is an expected consequence of the merging methodology: by relying on binary commercial ties between supplier and customer firms, the projection inherently assumes potential relationships among all combinations of products manufactured by the two entities. Consequently, the Cartesian product of highly diversified corporate portfolios inevitably generates "ghost links"—artificial connections that mathematically emerge from the construction process but lack real-world economic grounding.

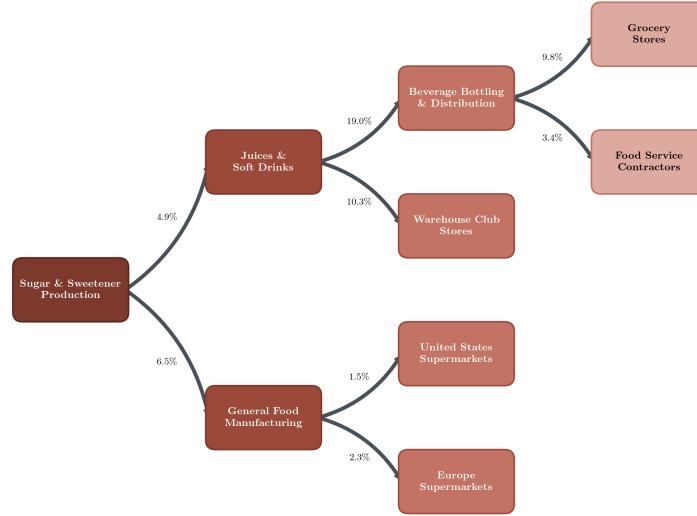
5.1. Qualitative Inspection of the Raw Value Chain

A preliminary manual examination of the network's specific connections provides immediate and concrete evidence of this structural noise. For instance, the raw network infers a commercial dependency between "Motion Picture and Television Distribution" and "Cattle Ranching and Farming", as well as between "Handheld and Smart Phone Games Software" and "Other Bacterial Infections Biopharmaceuticals". These spurious connections likely stem from minor overlapping revenue streams within highly diversified corporate portfolios. The probabilistic weighting mechanism partially neutralizes this noise: both of these economically absurd connections carry an infinitesimal normalized link weight of approximately 0.0001%.

However, structural anomalies can also manifest among strong links with highly significant weights, driven by the presence of colossal multinational conglomerates or massive investment funds. For example, the network reveals a strong dependency (normalized weight of 20.4%) between "Healthcare Facility Business Management" and "Conventional Engine Car Manufacturers", and an even more striking link (47%) between "China Video and Television Services" and "Dairy Products Production". Rather than algorithmic errors, these macro-



(a) Heavy-manufacturing pathway: Bauxite



(b) Agro-food distribution pathway: Sugar

Figure 4: **Hierarchical structure of empirical value chain pathways.** The figure displays two illustrative subgraphs extracted from the 2018 value chain. Both trees highlight the directional flow of goods across progressive economic stages: originating from raw materials or base ingredients, passing through intermediate manufacturing and processing, reaching final product assembly or large-scale distribution, and flowing into retail and consumer services. Panel (a) traces a heavy-manufacturing industrial pathway starting from "Bauxite Ore Mining", whereas panel (b) depicts an agro-food distribution chain originating from "Sugar and Sweetener Production". The percentages on the directed edges denote the normalized link weights, quantifying the strength of the inferred commercial dependencies between consecutive sectors. Note that these subgraphs serve as illustrative projections: within the macroscopic multi-sector network, source nodes possess multiple inbound supply dependencies and terminal nodes exhibit non-zero outbound links, reflecting the continuous, interconnected nature of the global economy.

scopic anomalies reflect the extreme diversification of modern corporate giants—such as massive holding companies that simultaneously operate heavy automotive manufacturing divisions alongside corporate healthcare networks, or Asian conglomerates merging media and entertainment with food and agricultural production. Because the supply chain dataset maps these conglomerates as single commercial entities, the resulting projection structurally bridges entirely unrelated sectors.

Despite these highlighted anomalies, the network also contains highly coherent commercial pathways, such as the ones schematized in Figure 4. The figure illustrates two distinct paths originating respectively from "Bauxite Ore Mining" and "Sugar and Sweetener Production", which evolve through additional intermediate processing stages before reaching their functional endpoints. However, these diagrams represent isolated subgraphs extracted for visual clarity. As evidenced by the fractional edge weights, these specific pathways are embedded within a much denser, highly interconnected network, where each node simultaneously maintains hundreds of other intersecting commercial ties.

When examining the network’s extremities—namely, the absolute basic inputs (nodes with zero in-degree, $k_{in} = 0$) and final outputs (nodes with zero out-degree, $k_{out} = 0$)—the topology reveals a mix of economically coherent sectors and structural anomalies. For instance, the source nodes correctly identify plausible primary inputs such as "Oilseed and Grain Farming", "North America Lithium Ore Mining", and "Geothermal Energy Equipment Manufacturing". However, they also include numerous service-oriented nodes like "United States Wireless Business Data Services" and "Environmental Restoration Services", which theoretically should not act as absolute starting points in a physical value chain. Similarly, among the terminal products, intuitive endpoints like "Beverage and Specialty Foods Stores", "Diabetes Diagnostic Testing Devices", and "Education Finance" appear alongside counterintuitive leaf nodes such as "Cannabis Farming" and "Diversified Lithium Ore Mining", which logically ought to feed into further downstream manufacturing rather than acting as final economic sinks.

This qualitative inspection suggests that while physical goods and manufacturing sectors generally follow a directed, coherent sequence—albeit with inherent noise—services and non-physical commodities exhibit a highly circular and nested topol-

ogy. This behavior can be explained by the real-world structure of large investment funds and multinational conglomerates that cross-invest across disparate fields, thereby entangling the economic network. The structural impact of corporate diversification becomes evident when analyzing the major network hubs. Ranked by their in-degree and out-degree, the most densely connected nodes correspond to broad service categories and corporate accounting structures, such as "General or Multi-Industry Revenue" ($k_{in} = 1375$, $k_{out} = 1385$), "Commercial Property Owners" ($k_{in} = 1217$, $k_{out} = 1150$), and "General Information Technology (IT) Consulting" ($k_{in} = 1137$, $k_{out} = 998$).

Furthermore, the RBICS taxonomy incorporates residual macro-categories (such as "Other" and "Non-Corporate", as detailed in Appendix A), which are designed to aggregate products and services lacking a more specific classification. Because of this hierarchical structure, the raw value chain includes overly broad and analytically uninformative nodes, such as "Other Infrastructure Construction" or "Other Petroleum Products". This taxonomic feature, combined with overly specific geographic subdivisions (e.g., explicitly defining a sector as "Asia (Excluding China) Wireless Services"), introduces additional structural noise into the network. These classification artifacts inevitably blur the economic boundaries between sectors, further complicating the extraction of pure commercial pathways from the raw dataset.

5.2. Statistical Characterization of the Value Chain

Having framed the empirical value chain as a complex network, we can leverage tools from network science to quantitatively characterize its topological properties. Figure 5 displays the absolute frequency counts of the in-degree (Figure 5a) and out-degree (Figure 5b) distributions. Both histograms are shown on a log-log scale and exhibit highly similar profiles, indicating that the system displays structural symmetry regarding its inbound and outbound connectivity; therefore, for the sake of simplicity, we will refer to both k_{in} and k_{out} simply as k when discussing their shared statistical behaviors.

The distributions reveal three distinct structural regimes. First, an ascending phase in the log-log space spanning from $k = 10^0$ to $k \approx 10^2$ illustrates the progressive accumulation of nodes toward the network’s modal connectivity. This is followed by a broad, densely populated plateau between $k = 10^2$

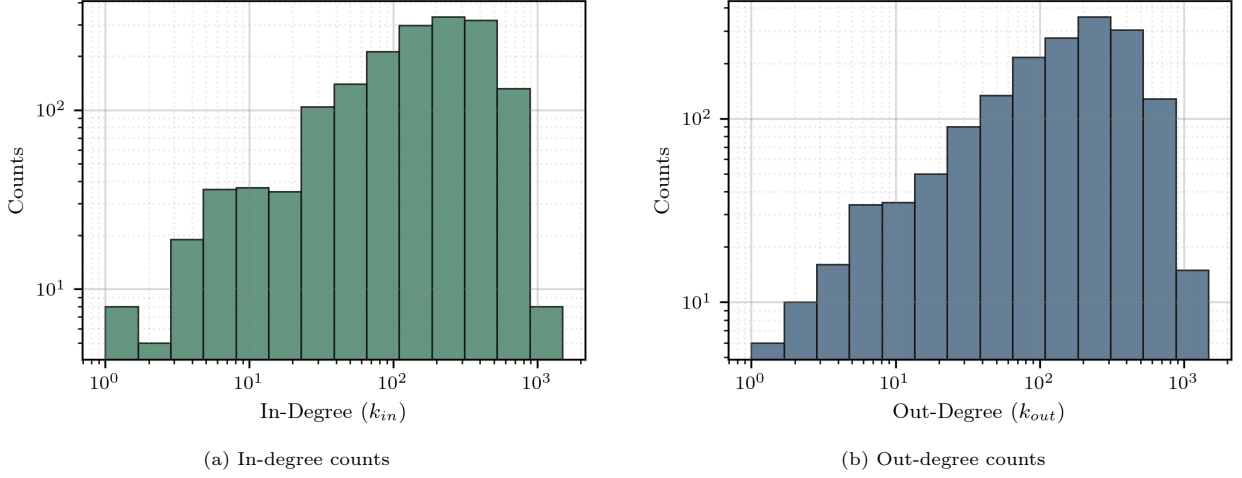


Figure 5: **Empirical degree distributions of the 2018 value chain network.** The figure displays the absolute frequency counts of (a) in-degrees (k_{in}) and (b) out-degrees (k_{out}) for the product nodes. Both distributions are computed using logarithmic binning and presented on a log-log scale, spanning from 10^0 to over 10^3 . The highly right-skewed profiles reveal a complex topology typical of large economic networks: while the vast majority of products maintain a limited number of specialized commercial dependencies, a small fraction of macroscopic hubs exhibit extreme degrees exceeding 10^3 .

and $k \approx 5 \times 10^2$, and finally a sharp cutoff beyond $k \approx 5 \times 10^2$. In economic terms, this topology implies that while a significant portion of products ($\approx 34\%$) maintains a restricted number of direct dependencies ($k < 10^2$), the absolute majority of the network (65.6%) consists of medium-to-high degree nodes ($10^2 \leq k < 10^3$). Conversely, macroscopic hubs with extreme connectivity ($k \geq 10^3$) represent an entirely marginal fraction of the system.

Shifting the focus to link weights, Figure 6 illustrates the absolute frequencies and the empirical Cumulative Distribution Function (CDF) of the transition probabilities w (i.e., normalized link weights). The frequency histogram (Figure 6a) yields a remarkably broad profile spanning over ten orders of magnitude. On a log-log scale, the distribution exhibits an accumulation of mass from $w \approx 10^{-11}$ to $w \approx 10^{-4}$, culminating in a smooth, densely populated peak in the interval between 10^{-4} and 10^{-2} . This massive aggregation of weak ties is rigorously corroborated by the exact CDF (Figure 6b), which provides robust evidence of the network's highly right-skewed dynamics. Beyond the $\approx 10^{-2}$ threshold, the decay rate steepens precipitously, driving a sharp structural cutoff for the highest transition probabilities.

From an economic perspective, this pronounced skewness confirms that the network is flooded by in-

finitesimally weak connections (23% of links exhibit $w \leq 10^{-4}$) alongside a massive core of intermediate dependencies (68.21% falling between 10^{-4} and 10^{-2}). In contrast, edges carrying a heavy transitional weight ($w \geq 10^{-2}$) constitute a remarkably small minority of the total value chain (merely 8.75%). This accumulation of weak ties drastically dilutes the network's overall mean weight to just $\bar{w} = 0.0044\%$.

Given the overwhelming presence of topological noise in the single-year snapshot, a temporal analysis becomes essential to investigate whether a stable structural core persists over time.

6. Temporal Persistence Analysis (2018-2023)

Until now, our analysis has focused exclusively on the 2018 value chain. Following the same methodology described in Section 4, we constructed the equivalent value chain networks for the consecutive years spanning from 2019 to 2023. Having generated this sequence of networks, it is crucial to compare their topological structures to detect which links are persistent over time. Identifying this persistent core could allow us to distinguish genuine, structurally carrying commercial relationships from transient or occasional economic noise, ultimately

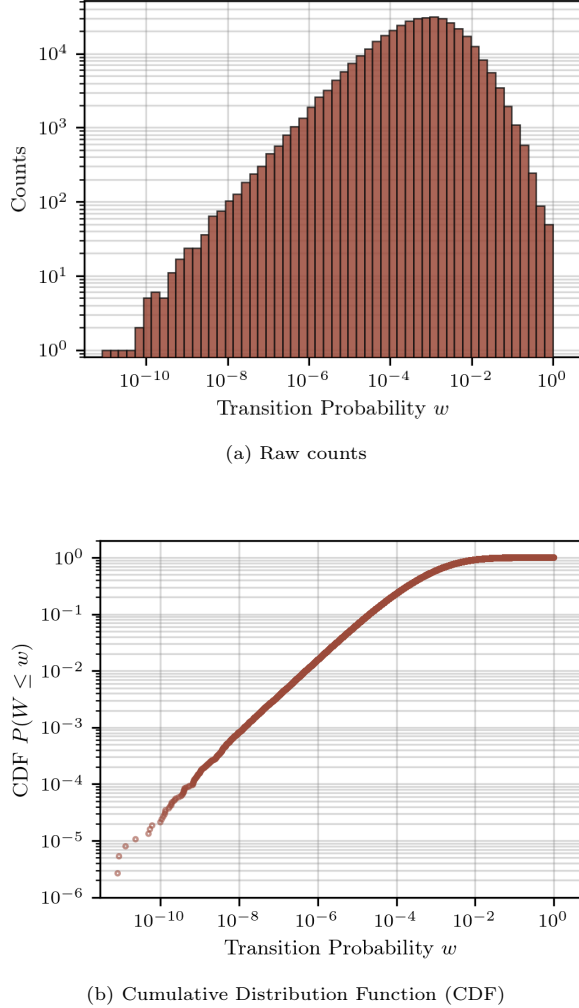


Figure 6: Empirical distributions of transition probabilities. The figure displays the topological distribution of the normalized link weights w characterizing the 2018 value chain. Panel (a) shows the absolute frequency counts computed via logarithmic binning, while panel (b) illustrates the exact Cumulative Distribution Function $P(W \leq w)$. Both profiles are presented on a log-log scale and convey complementary structural information: the system spans over ten orders of magnitude, with panel (a) highlighting the massive accumulation of intermediate-weight connections and panel (b) revealing the cumulative distribution across the spectrum. The smooth growth visible in the CDF demonstrates that infinitesimal and weak connections make up a substantial baseline fraction, whereas genuine, high-weight commercial dependencies concentrate heavily toward the upper bound of the distribution.

yielding a cleaner and more robust representation of the global value chain.

6.1. Macroscopic Structural Stability

To assess the temporal evolution of the network at a macroscopic level, Figure 7 displays the structural comparison among all the constructed annual networks through the computation of the pairwise Jaccard similarity index, which is formally defined as:

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|} \quad (3)$$

where A and B represent the edge sets of the two compared years, $|A \cap B|$ denotes the number of persistent links (the intersection), and $|A \cup B|$ is the total number of links existing across both networks (the union).

The matrix reveals a highly consistent temporal dynamic. For any adjacent year pair (i.e., year-over-year transitions), the network exhibits substantial topological persistence, maintaining a stable Jaccard index ranging between 0.54 and 0.65. This uniformity indicates the absence of sudden topological shocks or systemic disruptions during the observed period; rather, the network experiences a physiological structural turnover (entries and exits of marginal ties) that leaves its core economic backbone intact. As the temporal distance increases, the structural overlap monotonically and smoothly decreases. For non-consecutive transitions (spanning 2 to 5 years), the similarity decays, reaching its minimum at $J = 0.35$ for the 2018-2023 span. In the context of large-scale complex networks, a 35% structural overlap evaluated across a half-decade still indicates a massive and highly solid persistent core. Furthermore, the symmetric and smooth nature of this decay confirms that annual changes accumulate gradually without abrupt systemic shifts. However, it is crucial to acknowledge that over an extended five-year window, the cumulative loss of edges inevitably captures genuine macroeconomic market variations and structural realignments, meaning that the pruned connections cannot be exclusively attributed to the dissipation of transient economic noise.

6.2. Weight Distributions of Persistent and Transient Links

Building upon this macroscopic evidence of structural persistence, we now shift our focus to the mi-

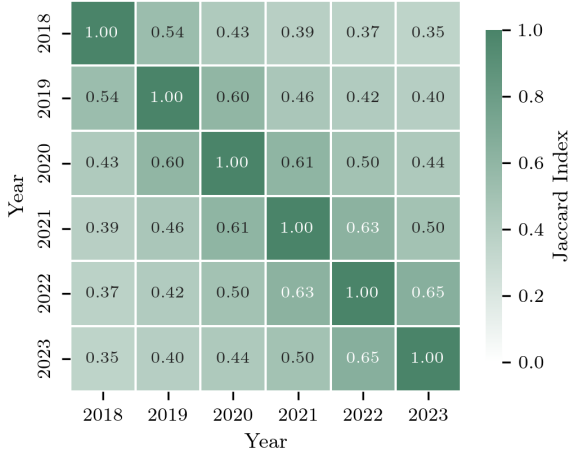


Figure 7: **Temporal stability of the value chain network (2018-2023).** The heatmap displays the pairwise Jaccard similarity index across all observed years, evaluating the structural overlap of the network’s active edges. The index ranges from 0 (no shared connections) to 1 (perfectly identical topologies, as strictly observed on the main diagonal). The matrix demonstrates a coherent structural persistence: the similarity smoothly and monotonically decays as the temporal distance between any compared pair of years increases.

croscopic level, analyzing the topological weights to investigate whether distinct link categories exhibit different strength profiles. From an economic perspective, we hypothesize that the persistent backbone is sustained by structurally robust edges with higher transitional weights, whereas occasional or transient connections (appearing sporadically across the time frame) represent weaker relationships or structural noise. To quantitatively test this hypothesis, we compare two annual value chain networks, denoted as A and B , and classify their edges into three distinct species: "lost links" (present in A but extinct in B), "new links" (emerging in B but absent in A), and "persistent links" (the intersection, present in both A and B).

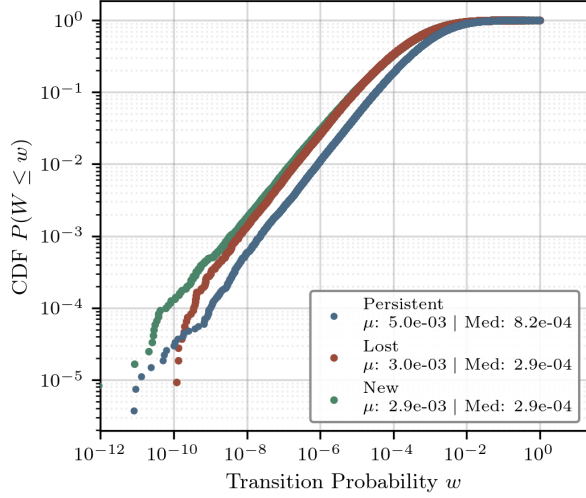
By classifying the edges for specific year-pairs, we can compute and compare their weight distributions to determine if these three species are characterized by inherently different statistical properties.

Figure 8 illustrates the Cumulative Distribution Functions (CDFs) of the transition probabilities w for the three link species, comparing the 2018-2019 period and the 2018-2023 span on a log-log scale. For adjacent years (Figure 8a), the plot reveals a clear topological separation: the curve of the persistent links (blue dots) is distinctly shifted toward

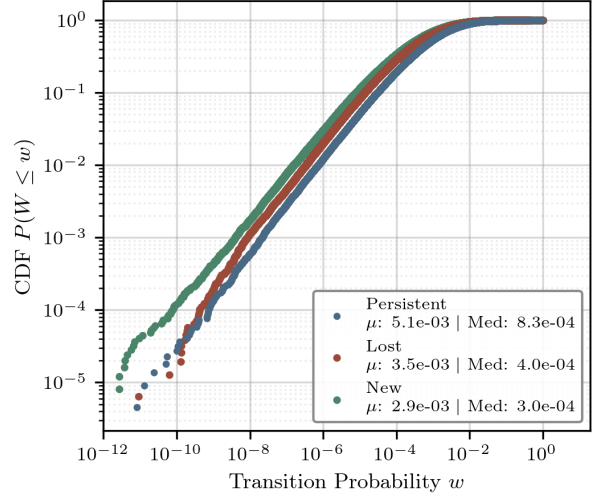
higher weights compared to the new and lost links (green and red dots, respectively). The legend confirms this visual gap: the mean weight of the persistent distribution is nearly twice that of the other two, while the median is approximately three times higher. Conversely, new and lost links share almost identical statistical behaviors, being generally restricted to significantly lower transitional probabilities.

This structural separation remains evident, albeit slightly attenuated, in the 2018-2023 comparison (Figure 8b). Over a five-year gap, the three distributions remain distinguishable, but the distance between persistent and transient edges narrows. This behavior suggests that over longer temporal windows, structural turnover inherently increases, capturing more significant macroeconomic shifts rather than mere noise. This interpretation is corroborated by the network’s survival rates: while 71.5% of the active links survived the year-over-year transition from 2018 to 2019, only 58.6% persisted over the entire half-decade from 2018 to 2023. Remarkably, despite this increased turnover, the mean weight of the persistent links remains virtually constant across both temporal pairs, confirming the resilience of the network’s heavy-weight linkages.

As further macroscopic evidence of this temporal dynamics, the extraction of the 5-year persistent network reveals a dramatic drop in density compared to the raw 2018 topology, falling from $\rho = 12.9\%$ to $\rho = 5.86\%$. This stable core comprises 148,966 persistent edges among 1,595 nodes; notably, over the five-year span, only 185 nodes disappeared from the system, whereas nearly 60% of the original edges were pruned. This significant structural reduction confirms that temporal evolution naturally filters out a massive amount of transient economic noise. However, a 5.86% density remains structurally too high for a pure economic network. A qualitative inspection of the absolute persistent dependencies (links maintaining a transition probability of $w = 1.0$ across the half-decade) corroborates this limitation. While some pathways are logically sound (e.g., "Natural Gas Compression Equipment" $\rightarrow$ "MENA Fossil Fuel Exploration/Production", "Boat and Recreational Vehicle Sales" $\rightarrow$ "Travel Agencies", or "Diversified Seafood Production" $\rightarrow$ "Warehouse Club Stores and Superstores"), highly stable but entirely spurious connections also persist. Counterintuitive dependencies, such as those between "Funeral and Cemetery Services" and "Burger Restaurants", or "Cotton Farm-



(a) CDF comparison (2018-2019)



(b) CDF comparison (2018-2023)

Figure 8: **Weight distributions by link dynamics.** Empirical CDFs of transition probabilities for persistent, lost, and new links, evaluated for adjacent years (a) and over a five-year gap (b).

ing” and “Sweets and Snacks Production”, clearly illustrate this flaw. These enduring artifacts perfectly demonstrate that, although temporal stability strongly suggests the existence of an underlying economic backbone, time-based filtering alone cannot dismantle the artificial ties generated by highly diversified corporate conglomerates.

7. Discussion and Conclusions

This report presented an exploratory topological analysis of the global value chain, constructed by integrating the FactSet RBICS Revenue and Supply Chain Relationships databases. Following a rigorous data pre-processing and cleaning phase, the empirical firm-level interactions were successfully projected into a granular product space. However, as analyzed in Section 4, both qualitative and statistical assessments of the resulting raw network for the baseline year 2018 exposed significant structural limitations.

Qualitatively, the network exhibited an anomalously high macroscopic density ($\rho = 12.9\%$). The projection mechanism, which relies on the Cartesian product of diversified corporate portfolios, inevitably generated a densely entangled topology severely polluted by structural noise and “ghost links”. This artifact was concretely corroborated

by the emergence of economically counterintuitive dependencies, such as the spurious tie between “Motion Picture and Television Distribution” and “Cattle Ranching and Farming”. Furthermore, this topological entanglement is exacerbated by the RBICS taxonomy itself; the presence of residual macro-categories (such as “Other” or “Non-Corporate”) inherently blurs sectoral boundaries and injects additional uninformative dependencies into the system.

The subsequent statistical characterization rigorously confirmed this noisy landscape. Both frequency counts and Cumulative Distribution Functions (CDFs) demonstrate that transition probabilities span over ten orders of magnitude, exhibiting a broadly distributed, right-skewed profile bounded by a sharp structural cutoff. Concurrently, the degree distribution reveals that a decisive majority of product nodes (65.6%) is concentrated within a dense connectivity range ($10^2 \leq k < 10^3$). This topological configuration highlights the pervasive presence of noise: an extensive fraction of the network (23% of links) consists of infinitesimally weak connections ($w \leq 10^{-4}$), while the system is overwhelmingly dominated by these faint and intermediate dependencies. This massive accumulation collectively dampens the overall mean weight to a marginal $\hat{w} = 0.0044$, effectively obscuring the

small minority of genuine, strong commercial linkages.

Despite this overwhelming presence of structural noise, the temporal analysis presented in Section 6 provided crucial insights into the network’s dynamics, strongly suggesting the existence of a solid structural backbone. By evaluating the network’s evolution from 2018 to 2023, the pairwise Jaccard similarity index revealed a coherent structural persistence. Specifically, 58.6% of the initial links persisted across the five-year span, maintaining a significant topological overlap and effectively reducing the network’s macroscopic density (from $\rho = 12.9\%$ to $\rho = 5.86\%$). Furthermore, a microscopic comparison of link dynamics demonstrated a clear statistical separation: the CDF of persistent edges is distinctly shifted toward higher transition probabilities compared to lost or newly formed connections. This observation supports the hypothesis that while transient macroeconomic noise likely drives a high turnover among weaker links, the underlying economic core tends to be characterized by highly stable, heavier-weight dependencies—even though genuine weak ties naturally remain an integral part of the real market structure.

To rigorously isolate this true topological backbone, it is necessary to systematically filter out the background noise. However, relying on arbitrary deterministic thresholds (e.g., deleting all links below a certain numerical weight) is analytically inadequate, due to the lack of a statistical criterion. Given the multiscale nature of the economic system, a uniform cut-off would inevitably destroy functionally significant weak ties while preserving massive yet spurious connections generated by highly diversified corporate conglomerates. Furthermore, relying solely on a temporal persistence filter would prove equally ineffective: because stable commercial ties between large conglomerates can produce enduring spurious projections—such as the counter-intuitive dependency between “Funeral and Cemetery Services” and “Burger Restaurants”—a time-based filter alone cannot guarantee the removal of structurally persistent ghost links.

Ultimately, these preliminary findings demonstrate that the raw value chain is too heavily obscured by topological artifacts to be analyzed directly. Instead, as anticipated in the Introduction (Section 1), the natural methodological continuation of this exploratory work requires the implementation of tools from network statistical mechanics. Specifically, the next phase of this re-

search will focus on the application of a maximum-entropy null model (Direct Bipartite Configuration Model). Rather than applying fixed thresholds, this approach generates an ensemble of randomized, synthetic supply chains that preserve fundamental topological constraints (such as the node degree sequence) while destroying structural correlations. By comparing empirical transition probabilities against this randomized ensemble, it will become possible to systematically evaluate the statistical significance of each specific tie, allowing for the extraction of a validated network and the unmasking of the pure, functional backbone of the global economy.

A. RBICS Taxonomy Overview

The FactSet Revere Business Industry Classification System (RBICS) provides a granular, multidimensional framework for categorizing global companies based on their precise product lines and services. As detailed in the official methodology guidelines [3], RBICS distinguishes itself from traditional classification schemes through its rigorous bottom-up construction.

While the top tiers (Levels 1 to 3) are market-defined to reflect macroeconomic views of stock comovement, the bottom tiers (Levels 4 to 6) are strictly product-based, mapping the actual goods and services exchanged in the market [3]. Table A.1 summarizes the core architectural features of the dataset.

The 12 Anchor Industries (Level 1)

The 14x6 matrix is rooted in 14 macro-sectors at Level 1 (L1). Among these, 12 act as the fundamental “anchor industries” that categorize the bulk of global economic activities, while the remaining two (Other and Non-Corporate) accommodate unclassified entities or financial vehicles. Table A.2 provides the official descriptions for these top-level sectors.

Table A.1: RBICS Features Summary. Data sourced from the official FactSet RBICS Methodology Guide [3].

Feature	FactSet RBICS Description
Taxonomy Structure	14 industries (12 anchor industries and 2 specialty industries). 6 levels of depth with increasing granularity. Stable structure over years.
Taxonomy Construction	Built from bottom-up by grouping products and services. Top levels 1 to 3 are market-defined (macro view of stock co-movement). Bottom levels 4 to 6 are product-based.
Data Sources	Primary sources directly disclosed by companies (regulatory filings, investor reports, websites, and press releases).
Coverage	Approximately 48,000 global companies.
History from	RBICS with Revenue: US from 2012; Non-US from 2014.

Table A.2: Descriptions of the top level 14 Economy sectors (L1). Data sourced from the official FactSet RBICS Methodology Guide [3].

ID	NAME	DESCRIPTION
10	Business Services	Services targeted toward businesses, including administrative, support, janitorial, and professional services.
15	Consumer Services	Services targeted toward individuals, including accommodation; food and beverage retail; gaming, arts, entertainment and recreation; and television, radio, film, and print media.
20	Consumer Cyclical	Products targeted toward individual or household use, including apparel, toys, school and art supplies, and electronics; motor vehicle sales and rental, and automotive parts and services; building materials, garden supplies, furniture, appliances, cabinetry, window treatments, and carpets.
25	Energy	Oil and gas exploration and production, pipeline transportation, refineries, and oil and gas equipment and services; leasing, mining and processing of coal and coke; uranium, radium, and vanadium mining.
30	Finance	Financial products and services offered by institutions involved in banking, insurance, investment, specialty finance, and real estate.
35	Healthcare	Products and services that are designed, developed, and utilized in the promotion of health and well-being, including medical services, health plans, medical devices, and biopharmaceuticals.
40	Industrials	Products and services for industrial use or with applications in aero-space, defense or security; transportation, construction, and related infrastructure; or farming, including equipment and machinery manufacture, wholesale, rental, and distribution and related support activities.
45	Non-Energy Materials	Basic and intermediate material products, including non-energy mining; forestry, timber logging, and lumber production; and chemical, plastic, paper, metal, and textile manufacturing.
50	Consumer Non-Cyclicals	Products targeted toward individual and consumer needs, including groceries, beverages, health and personal care items, kitchenware, decorative items, and household cleaning products.
55	Technology	Semiconductor, electronic, and optics based products and related software and services that directly or indirectly facilitate the creation, transfer, storage, manipulation, or interpretation of data, audio, and video.
60	Telecommunications	Services designed to promote or enhance transmission of voice, data, and video over various communications mediums, including cable, satellite, terrestrial-based wireless, and wireline mediums.
65	Utilities	Gas, electricity, and water services delivered directly to residential and commercial users.
90	Other	Revenue identified as being corporate or otherwise not directly derived from normal production operations; whose product allocation is not disclosed; or which is identified as being generally distributed or derived from a multiple industries with no one source predominating.
95	Non-Corporate	Government entities at all levels from local to supranational, and their agencies and departments; membership organizations; and trusts, investment vehicles, and financial non-physical entities.

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